

# Review D — Macroeconomic Determinants and Trade Openness: What Moves Market Power and Misallocation?

Laxmikanta Sarangi

Department of Humanities and Social Sciences, IIT Roorkee

July 2026

## Abstract

Reviews A–C established how markups, markdowns and misallocation are measured and how they interact. This review asks what *determines* them: trade policy, the exchange rate, and the macroeconomic environment. Four literatures are surveyed equation-by-equation: (i) the theory of pro-competitive trade effects, from the Lerner symmetry of protection through Melitz–Ottaviano variable markups and Atkeson–Burstein pricing-to-market; (ii) the empirical trade-liberalisation–markup literature, from the first-generation developing-country studies through the watershed of De Loecker, Goldberg, Khandelwal and Pavcnik (2016), whose input-cost channel reversed the expected sign for India; (iii) exchange-rate pass-through as a markup phenomenon, and the nascent literature connecting trade to labour-market power; and (iv) the macro determinants of misallocation — cyclical, credit conditions, and development — together with the political economy of why distortions persist, culminating in Hsieh, Klenow, Kyei Manu and Rockall (2026). A final section reviews the econometric designs available for a single-country panel — exposure/shift-share identification, and the panel VAR — and states the research gap Chapter 6 fills: no study jointly estimates the response of markups, markdowns and misallocation to tariffs and exchange-rate exposure on modern Indian plant data.

## 1 Introduction and scope

---

The thesis’s first two objectives measure product- and labour-market power (Review A) and misallocation (Review B) in Indian organised manufacturing, and their linkage (Review C). The third objective asks the determination question: when tariffs rise, when the rupee appreciates, when a state grows, what happens to the markup  $\mu$ , the markdown  $\psi$ , and the within-industry dispersion of revenue productivity? This review organises the answer-relevant literature in four passes — theory (§2), trade–markup evidence (§§3–4), exchange rates and factor markets (§§5–6), and macro–misallocation with its political economy (§§7–8) — and closes with econometric designs and the gap (§9). Throughout, notation follows Reviews A–B:  $\mu_{it} = \theta_{it}^M / \alpha_{it}^M$  is the production-

approach markup;  $\psi_{it} = (\theta_{it}^L / \alpha_{it}^L) / \mu_{it}$  the markdown, with  $\psi > 1$  indicating monopsony; and  $\text{var}(\ln \text{TFPR})$  the misallocation statistic.

## 2 Theory: why openness should discipline markups — and why it may not

---

### 2.1 The market-size and substitution channels

In monopolistic competition with CES demand, the markup is pinned at  $\sigma/(\sigma - 1)$  and trade cannot move it: pro-competitive effects require *variable* markups. Melitz & Ottaviano (2008) deliver them with linear demand. Firm  $i$ 's optimal price over marginal cost  $c_i$  satisfies

$$p_i = \frac{1}{2} (c_i + c_D), \quad \mu_i \equiv \frac{p_i}{c_i} = \frac{1}{2} \left( 1 + \frac{c_D}{c_i} \right), \quad (1)$$

where  $c_D$  is the domestic cost cutoff for survival. Opening to trade lowers  $c_D$  (tougher selection), compressing every survivor's markup: the **pro-competitive effect**. Two corollaries matter empirically. First, the effect is a *level* effect on the distribution, strongest for high-markup (low- $c_i$ ) firms; second, it operates through import competition in the *output* market, so it should load on output tariffs and on import-exposure interactions — exactly the objects Chapter 6 constructs.

### 2.2 Pricing-to-market and the markup elasticity

Atkeson & Burstein (2008) generate variable markups from nested CES with oligopoly: firm  $i$ 's markup depends on its market share  $s_i$ ,

$$\mu_i = \frac{\varepsilon(s_i)}{\varepsilon(s_i) - 1}, \quad \varepsilon(s_i) = \left[ s_i \frac{1}{\eta} + (1 - s_i) \frac{1}{\rho} \right]^{-1}, \quad (2)$$

with  $\rho > \eta$  (within-sector substitution exceeds cross-sector). Because  $\partial \mu_i / \partial s_i > 0$ , shocks that reallocate shares — tariffs, exchange rates — move markups mechanically; and because exporters absorb part of a cost shock in the markup, pass-through is incomplete (§5). The *markup elasticity*  $\Gamma_i \equiv \partial \ln \mu_i / \partial \ln s_i$  is the sufficient statistic: the superelasticity estimated in this thesis's Chapter 5 (0.183 for India) is its demand-side counterpart.

### 2.3 Lerner symmetry and the input-cost channel

Lerner (1936) taught that an import tariff is an export tax: protection raises the domestic relative price of importables *and* the cost of imported inputs. For measured markups the second margin is decisive. Writing marginal cost  $c_{it} = c(w_t, p_{it}^M(\tau_{st}^{inp}))$ , a tariff cut lowers input prices; if pass-through of cost to price is incomplete ( $\beta < 1$  in  $\Delta \ln p = \beta \Delta \ln c$ ), the measured markup *rises* with liberalisation:

$$\Delta \ln \mu_{it} = \Delta \ln p_{it} - \Delta \ln c_{it} = -(1 - \beta) \Delta \ln c_{it} > 0 \quad \text{when } \Delta \ln c_{it} < 0. \quad (3)$$

Equation (3) is the reading key for the Indian evidence in §4 — and, run in reverse, for the protection episode this thesis studies: tariff *increases* raise input costs and compress measured markups.

### 3 First-generation evidence: liberalisation disciplines margins

---

The early plant-data literature regressed price–cost margins on openness. [Levinsohn \(1993\)](#), using Turkey’s 1984 liberalisation, and [Harrison \(1994\)](#), using Côte d’Ivoire’s 1985 reform, found margins falling in liberalised sectors — the “imports-as-market-discipline” hypothesis. For India, [Krishna & Mitra \(1998\)](#) exploited the 1991 reform:

$$\text{margin}_{jt} = \alpha_j + \gamma_t + \delta \text{Reform}_t \times \text{Ind}_j + \varepsilon_{jt}, \quad (4)$$

finding markups falling and productivity growth rising after 1991. These studies share two limitations that Review A’s toolkit later fixed: margins were accounting ratios, not estimated markups; and tariff endogeneity was addressed only by the reform’s suddenness. [Topalova \(2010\)](#) and [Topalova & Khandelwal \(2011\)](#) sharpened identification for India using the externally-imposed uniformity of the 1991–97 tariff cuts, tariffs at the industry  $\times$  year level, and TFP outcomes: lower output tariffs raised competitive pressure, and lower *input* tariffs raised firm productivity via cheaper, better inputs — the input channel that dominates in §4.

### 4 The watershed: De Loecker, Goldberg, Khandelwal and Pavcnik (2016)

---

DLGKP combine the production-approach markup (Review A) with plant-level *prices* from Indian Prowess data, decomposing prices into marginal costs and markups:

$$\ln p_{ijt} = \ln \mu_{ijt} + \ln c_{ijt}, \quad \begin{cases} \Delta \tau_{st}^{\text{out}} \downarrow \Rightarrow p \downarrow \text{ (competition)} \\ \Delta \tau_{st}^{\text{inp}} \downarrow \Rightarrow c \downarrow \downarrow \text{ (input costs)}. \end{cases} \quad (5)$$

Their central estimates: input-tariff cuts lowered marginal costs by far more than prices fell, so **markups rose** with liberalisation — factory-gate prices fell about 18%, but costs fell about 31%. The expected pro-competitive compression (from output tariffs) was present but swamped by incomplete cost pass-through (3). Three lessons for this thesis. (i) Output and input tariffs are different treatments and must be measured separately (Chapter 6 measures output tariffs; input tariffs via input–output weights are flagged as the natural extension). (ii) The sign of the markup response to protection is an empirical question, not a theoretical presumption. (iii) Identification came from industry  $\times$  year tariff variation with firm and year effects — the exposure design Chapter 6 adopts.

### 5 Exchange rates as competition policy: pass-through and markups

---

#### 5.1 Pass-through is a markup phenomenon

The exchange-rate pass-through (ERPT) regression,

$$\Delta \ln p_{it}^x = \beta \Delta \ln e_t + \mathbf{x}_{it}' \gamma + \varepsilon_{it}, \quad (6)$$

consistently finds  $\beta < 1$  (incomplete pass-through) for exporters’ destination prices ([Goldberg & Knetter, 1997](#)). Under constant markups  $\beta$  would be one; incompleteness *is*

markup adjustment. [Berman et al. \(2012\)](#) show high-performance French exporters absorb depreciations in their markups (raising them) rather than prices; [Amiti et al. \(2014\)](#) close the loop by showing that import-intensive exporters have doubly low pass-through — the marginal-cost and markup channels reinforce. The implication for a domestic-market study is symmetric: an *appreciation* lowers rivals’ import prices in the home market, and domestic firms in import-exposed industries absorb part of the squeeze in their markups. This is precisely the sign Chapter 6 finds for India: REER appreciation interacted with pre-determined import exposure compresses markups at both NIC-2 and NIC-3.

## 5.2 From national series to identification

Because the REER is a single national series, it is collinear with time effects; the literature identifies its effects through *exposure interactions* — e.g., import penetration or imported-input shares frozen at a base year. The generic estimating equation,

$$Y_{st} = \beta (\text{REER}_t - \overline{\text{REER}}) \times s_{s,0}^{\text{imp}} + \alpha_s + \delta_t + \varepsilon_{st}, \quad (7)$$

is a shift-share (Bartik) design: a common shock differencing across predetermined exposures. Its econometrics — what must be exogenous, and how to conduct inference — is the subject of §9.

## 6 Trade, macro conditions, and labour-market power

---

The markdown literature (Review A, §§5–6) is only beginning to meet the trade literature. [Morlacco \(2020\)](#) estimates buyer power of French importers over foreign suppliers using the two-flexible-input wedge — the same object as  $\psi$  — showing openness and factor-market power interact. [Rubens \(2023\)](#) shows oligopsony power distorts input allocation in Chinese tobacco, connecting markdowns directly to misallocation. For developing-country labour markets, [Amodio & de Roux \(2023\)](#) document substantial plant-level monopsony in Colombia with wedges rising after import competition intensifies, and [Brooks et al. \(2021\)](#) provide the India–China markdown benchmarks this thesis extends. The theoretical presumption is two-sided: protection props up product demand and therefore labour demand, eroding wage-setting power in thin markets; import competition does the opposite. Chapter 6’s finding that tariffs (at fine industry detail) *reduce* monopsony incidence, while appreciation weakly raises it, is — to the best of this review’s knowledge — the first plant-panel estimate of the trade–monopsony margin for India.

## 7 Macroeconomic determinants of misallocation

---

### 7.1 Cycles, credit and crises

Misallocation is not acyclical. Financial frictions concentrate its movement: [Gilchrist et al. \(2017\)](#) model dispersion in borrowing costs as an interest-rate wedge; credit crunches raise MRPK dispersion where collateral constraints bind. For India specifically, [Bau & Matray \(2023\)](#) show that easing foreign capital access (FDI liberalisation across industries and time) *reduced* capital misallocation — MRPK dispersion fell in treated industries — establishing that Indian misallocation responds to factor-market policy even

where it ignores trade policy. [Asker et al. \(2014\)](#) provide the permanent caution: volatility plus adjustment costs generates TFPR dispersion mechanically, so cyclical dispersion need not be distortion.

## 7.2 Development and the wedge distribution

Cross-country, dispersion falls with income ([Hsieh & Klenow, 2009](#); [Restuccia & Rogerson, 2017](#)). Within a developing country over time, the sign is unsettled: growth episodes attract heterogeneous entrants and expand constrained incumbents unevenly, so measured dispersion can *rise* along the transition path even as policy improves — the reading this thesis gives its state-level finding that Indian states growing faster than trend see TFPR dispersion fall by less. The HKKR result (§8) supplies the complementary political-economy reason why measured wedges persist.

## 8 Why distortions persist: political economy

---

[Grossman & Helpman \(1994\)](#) formalise protection-for-sale: tariffs respond to organised interests, so trade policy is an *outcome* of industry conditions — markups, employment, distress — as much as a cause. [Bown & Tovar \(2011\)](#) document exactly this for India: after the 1990s tariff reform, antidumping and safeguards partially restored protection where liberalisation bit hardest. The econometric implication is first-order: single-equation regressions of market power on tariffs face reverse causality, motivating (i) pre-determined exposure designs and (ii) explicit feedback modelling. Chapter 6’s panel VAR delivers the direct evidence — markups and misallocation Granger-cause tariffs in 2012–2022 India. [Hsieh et al. \(2026\)](#) close the circle at the level of welfare: removing India’s size-dependent wedges would raise output 1–4% but lower consumption-equivalent welfare ~9% because losses concentrate on low-skill workers and small-firm owners; majorities therefore rationally oppose the reform. Distortions persist because their removal is distributionally unattractive — protection persists for the same reason.

## 9 Econometric designs for a single-country panel, and the research gap

---

### 9.1 Exposure (shift-share) designs

Equation (7) generalises to any national shock  $g_t$  interacted with predetermined exposure  $s_{s,0}$ . Identification requires either exogenous shares ([Goldsmith-Pinkham et al., 2020](#)) or many quasi-random shocks ([Borusyak et al., 2022](#)); inference should cluster at the exposure level. In Chapter 6 the shares are 2012 import shares (predetermined by construction), the shocks are the REER path and industry tariffs, and year effects absorb all national aggregates — the design estimates *relative* responses, the honest object for a single country.

### 9.2 Panel vector autoregression

For dynamics and feedback with unit heterogeneity, the panel VAR of [Love & Zicchino \(2006\)](#) and [Abrigo & Love \(2016\)](#),

$$\mathbf{y}_{it} = \mathbf{A}(L) \mathbf{y}_{it-1} + \mathbf{u}_i + \mathbf{e}_{it}, \quad (8)$$

estimated by GMM with forward orthogonal deviations (Helmert transform) to remove  $\mathbf{u}_i$  without Nickell bias, delivers Granger tests and orthogonalised impulse responses. Two practice notes this thesis validates: common time effects must be removed first (year-demeaning) or trending panels violate stability; and lag order should be chosen by the Andrews–Lu criteria. With  $T = 11$  the horizon is short; IRFs are descriptive of predictive dynamics, not structural shocks.

### 9.3 The research gap

Across §§3–7 no study estimates, on modern Indian plant data, the joint response of the three thesis objects — markups, markdowns/monopsony, and misallocation — to trade policy and exchange-rate exposure, with the political-economy feedback made explicit. The 1990s liberalisation literature stops at productivity or (for DLGKP) markups, before markdowns were measurable; the markdown literature has not met Indian trade policy; the misallocation literature’s Indian branch (Bau–Matray; HKKR) studies capital-market and size-dependent policy, not the border. Chapter 6 fills this gap with an eleven-year industry and state panel carrying all three objects, a newly concorded NIC-2008 tariff panel (2011–2023), REER-exposure interactions, state accounts, and a stability-verified panel VAR — and finds protection compressing markups (the DLGKP cost channel in reverse), appreciation disciplining exposed industries, monopsony falling with protection, misallocation orthogonal to the border but rising with state growth, and tariffs responding to industry conditions.

## References

---

- Abrigo, M. R. M., & Love, I. (2016). Estimation of panel vector autoregression in Stata. *The Stata Journal*, 16(3), 778–804.
- Amiti, M., Itskhoki, O., & Konings, J. (2014). Importers, exporters, and exchange rate disconnect. *American Economic Review*, 104(7), 1942–1978.
- Amodio, F., & de Roux, N. (2023). Labor market power in developing countries: Evidence from Colombian plants. *Journal of Labor Economics* (forthcoming version; working paper 2021).
- Asker, J., Collard-Wexler, A., & De Loecker, J. (2014). Dynamic inputs and resource (mis)allocation. *Journal of Political Economy*, 122(5), 1013–1063.
- Atkeson, A., & Burstein, A. (2008). Pricing-to-market, trade costs, and international relative prices. *American Economic Review*, 98(5), 1998–2031.
- Bau, N., & Matray, A. (2023). Misallocation and capital market integration: Evidence from India. *Econometrica*, 91(1), 67–106.
- Berman, N., Martin, P., & Mayer, T. (2012). How do different exporters react to exchange rate changes? *Quarterly Journal of Economics*, 127(1), 437–492.
- Borusyak, K., Hull, P., & Jaravel, X. (2022). Quasi-experimental shift-share research designs. *Review of Economic Studies*, 89(1), 181–213.
- Bown, C. P., & Tovar, P. (2011). Trade liberalization, antidumping, and safeguards: Evidence from India’s tariff reform. *Journal of Development Economics*, 96(1), 115–125.
- Brooks, W. J., Kaboski, J. P., Li, Y. A., & Qian, W. (2021). Exploitation of labor?

- Classical monopsony power and labor's share. *Journal of Development Economics*, 150, 102627.
- De Loecker, J., Goldberg, P. K., Khandelwal, A. K., & Pavcnik, N. (2016). Prices, markups, and trade reform. *Econometrica*, 84(2), 445–510.
- Gilchrist, S., Sim, J. W., & Zakrajšek, E. (2017). Uncertainty, financial frictions, and investment dynamics. In *NBER Working Paper* No. 20038 (revised).
- Goldberg, P. K., & Knetter, M. M. (1997). Goods prices and exchange rates: What have we learned? *Journal of Economic Literature*, 35(3), 1243–1272.
- Goldsmith-Pinkham, P., Sorkin, I., & Swift, H. (2020). Bartik instruments: What, when, why, and how. *American Economic Review*, 110(8), 2586–2624.
- Grossman, G. M., & Helpman, E. (1994). Protection for sale. *American Economic Review*, 84(4), 833–850.
- Harrison, A. E. (1994). Productivity, imperfect competition and trade reform: Theory and evidence. *Journal of International Economics*, 36(1–2), 53–73.
- Hsieh, C.-T., & Klenow, P. J. (2009). Misallocation and manufacturing TFP in China and India. *Quarterly Journal of Economics*, 124(4), 1403–1448.
- Hsieh, C.-T., Klenow, P. J., Kyei Manu, E., & Rockall, E. (2026). Misallocation versus inequality. Working paper.
- Krishna, P., & Mitra, D. (1998). Trade liberalization, market discipline and productivity growth: New evidence from India. *Journal of Development Economics*, 56(2), 447–462.
- Lerner, A. P. (1936). The symmetry between import and export taxes. *Economica*, 3(11), 306–313.
- Levinsohn, J. (1993). Testing the imports-as-market-discipline hypothesis. *Journal of International Economics*, 35(1–2), 1–22.
- Love, I., & Zicchino, L. (2006). Financial development and dynamic investment behavior: Evidence from panel VAR. *The Quarterly Review of Economics and Finance*, 46(2), 190–210.
- Melitz, M. J., & Ottaviano, G. I. P. (2008). Market size, trade, and productivity. *Review of Economic Studies*, 75(1), 295–316.
- Morlacco, M. (2020). Market power in input markets: Theory and evidence from French manufacturing. Working paper, University of Southern California.
- Restuccia, D., & Rogerson, R. (2017). The causes of misallocation. *Journal of Economic Perspectives*, 31(3), 151–174.
- Rubens, M. (2023). Market structure, oligopsony power, and productivity. *American Economic Review*, 113(9), 2382–2410.
- Topalova, P. (2010). Factor immobility and regional impacts of trade liberalization: Evidence on poverty from India. *American Economic Journal: Applied Economics*, 2(4), 1–41.
- Topalova, P., & Khandelwal, A. (2011). Trade liberalization and firm productivity: The case of India. *Review of Economics and Statistics*, 93(3), 995–1009.